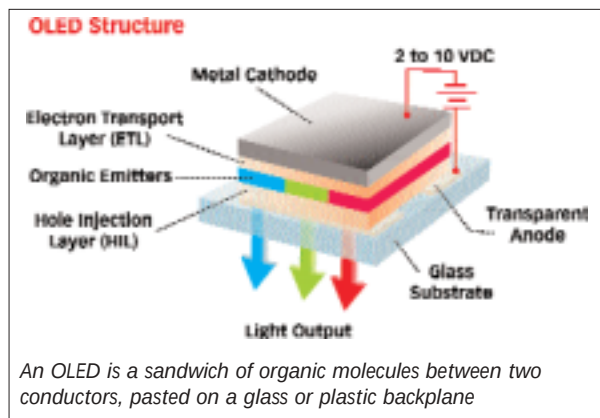




The EasyShare uses an OLED display instead of the conventional LCD

An LCD display has a backlight that is always on at full brightness and the liquid crystal is used as a polarising layer, to control the amount of light allowed to pass through. OLEDs on the other hand generate their own light and do not require a separate backlight layer. No backlight means one big layer eliminated from the flat panel display. Just this fact makes OLEDs economically attractive. Also, since the OLED only uses electricity in proportion to the amount of light to be generated, it saves on power. According to a Kodak technical paper, the elimination of the backlight and one sheet of glass combined with the thinner layers required for the organic emitter substance contribute to a much thinner display. An LCD active matrix display would be about 6 mm thick, while a comparable OLED active matrix display would only measure 1.6 mm thick.

But the goodness doesn't stop there. OLED displays have a much greater contrast range than LCDs, and approach the limits of the sensitivity of the human eye. This means that we can expect to see sharp, rich colours far superior to other conventional displays. Because there is no polarising layer in OLEDs, the viewing angle is far greater than LCDs. The polarising layer limits the angular range of light emission from the backlight. For example, picture a light in the back of a box. When you open the lid, you can see the light only from the range permitted by the box lid. However, the OLEDs themselves emit light right from the surface, like a light source on the lid itself.

Lighter, brighter and low-power requirements make OLEDs better suited to mobile applications. At the current state of technology, mass production of displays greater than 4 inches ** (is this right?) is unfeasible, though under laboratory conditions large screen displays

have been made. Already there are prototype displays available for licensing from some of the pioneers, and Kodak has gone ahead and put their display into production. The recently launched EasyShare LS633 digital camera uses an OLED display for the camera interface instead of an LCD.

Current fabrication technology limits OLEDs to glass substrates, but research is on for manufacturing OLEDs on flexible plastic backplanes as well. This means that the display can be bent, folded or rolled up and yet function perfectly. Better still, transparent OLEDs can be applied to windows or the windshield of a car to give a Heads-Up Display (HUD), similar to the ones used in fighter jets. Universal Display recently received a grant from the U.S military to continue research on laying down OLED displays on flexible metal foil. The applications are clear—a large, bright, low-power display that can be conveniently pocketed for mobility. On colour PDAs, the major battery drain is the huge backlit screen. With OLEDs consuming potentially half the power, you can expect significantly lighter devices that stay lit longer.

There are technical with OLEDs that are yet to be resolved. The most important one is that of the lifespan of the device. Researchers are facing trouble with blue light emitting materials. Kodak claims to have achieved a 10,000 hour lifetime for its blue organic materials, while hitting 40,000 hours for red

and green. For mobile applications where the screen is only used intermittently through the life of the device, these are approaching adequate figures, but for desktop PC applications, or for use in flat screen televisions, these lifetime figures have to be boosted much more.

In the early production models, the power consumption is comparable to LCDs, but this is expected to be lowered in later generations.

OLED displays are at a stage where they are just beginning to move into practical applications and there is tremendous interest in future developments possible with this technology.

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